Paper Code : 21312

F- 412

B.C.A. (Third Semester) Examination, 2024-25

(New Course)

Paper - BCA-302

Computer Organization

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Describe in detail constructing of Arithmetic Logic Unit (ALU)

   (b) What is Floating point arithmetic? Explain with example.

2. (a) Explain the difference between cache memory and virtual memory.

   (b) Explain Memory Hierarchy with example.

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3. (a) Explain segmented memory and paged segment memory.

   (b) Differentiate Magnetic Disk and Magnetic Tape.

4. (a) Explain arithmetic operation on binary operation with example.

   (b) Explain Index Addressing Mode with examples..

5. (a) Explain the DMA control with the help of block diagram.

   (b) Divide using division algorithm 10100011 by 1011

6. (a) Subtract 101 - 0.101 with 2's complement method.

   (b) Differentiate serial bus arbitration with parallel bus arbitration.

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7. (a) What is interrupt? Why it is used? explain with example.

(b) Differentiate between RISC and CISC

8. Write short notes on any four of the following.

(i) Static & Dynamic RAM

(ii) Instruction types.

(iii) Programmed I/O

(iv) High speed memories and its uses

(iv) Octal and Hexa decimal Number system.

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